

Common Sports Supplements

Creatine

Creatine is a naturally occurring compound involved in rapid energy production.

Muscle phosphocreatine helps regenerate ATP during short, intense efforts.

Evidence

Creatine is one of the most extensively studied sports supplements.

Creatine supplementation can improve performance in activities involving repeated high-intensity efforts and can support:

- strength gains;
- power;
- resistance-training performance;
- muscle gain when combined with appropriate training.

Creatine is less relevant as a direct performance enhancer for steady-state endurance activity.

Form

Creatine monohydrate is the most widely studied form.

More expensive forms have not consistently been shown to outperform creatine monohydrate.

Common Intake

A commonly studied maintenance intake in healthy adults is:

3-5 g per day.

Some protocols use a loading phase followed by maintenance intake.

Loading is not required. It mainly saturates muscle creatine stores more quickly.

Body Weight

Creatine can cause an initial increase in body weight partly because of increased water associated with muscle creatine storage.

This should not automatically be interpreted as body-fat gain.

Whey Protein

Whey protein is a food-derived protein supplement.

It can be useful when:

- normal meals do not provide enough protein;
- a portable protein source is needed;
- preparing a full meal is inconvenient.

Whey is not required for muscle growth if sufficient protein is already consumed from food.

Common forms include:

- whey concentrate;
- whey isolate;
- whey hydrolysate.

Casein Protein

Casein is a milk-derived protein that digests more slowly than whey.

It can be used as a convenient protein source, particularly when a person wants a slower-digesting protein food.

Casein is not required if total daily protein intake is already sufficient.

Plant Protein Powders

Common plant protein powders include:

- soy protein;
- pea protein;
- rice protein;
- blended plant proteins.

They can be useful for:

- vegans;
- people avoiding dairy;
- people who prefer plant-based protein sources.

Blended plant protein products can provide a broader amino-acid profile.

Caffeine

Caffeine is a stimulant.

It can:

- increase alertness;
- reduce perceived fatigue;
- improve performance in some exercise settings.

Sports-performance research commonly studies doses around:

2-6 mg/kg body weight

before exercise.

Higher doses do not necessarily improve performance and can increase adverse effects.

Possible side effects include:

- disturbed sleep;
- restlessness;
- anxiety;
- nausea;
- increased heart rate;
- palpitations in susceptible individuals.

Many authorities consider total daily caffeine intake around **400 mg or less** unlikely to cause dangerous effects in most healthy adults, although individual sensitivity can be much lower.

Pure powdered caffeine is dangerous because small measuring errors can cause very large doses.

Branched-Chain Amino Acids

BCAAs are:

- leucine;
- isoleucine;
- valine.

They are naturally present in protein-rich foods and complete protein supplements.

For people already consuming enough high-quality protein, current evidence does not show a strong additional benefit from BCAA supplements for:

- muscle growth;
- recovery;
- performance.

Complete protein sources provide BCAAs together with the other essential amino acids.

Essential Amino Acids

Essential amino acids cannot be produced in sufficient amounts by the body and must come from the diet.

Complete protein foods naturally provide EAAs.

EAA supplements can provide amino acids, but they are not usually necessary when total dietary protein is already sufficient.

Beta-Alanine

Beta-alanine increases muscle carnosine.

It may improve performance in some high-intensity exercise where muscular acidity contributes to fatigue.

Beta-alanine is not a fat burner.

A common harmless side effect is temporary tingling of the skin called paresthesia.

Citrulline

Citrulline and citrulline malate are commonly marketed for:

- blood flow;
- exercise performance;
- muscle "pump."

Research results are mixed.

Evidence is less established than for creatine or caffeine.

Electrolytes

Electrolytes include minerals such as:

- sodium;
- potassium;
- chloride;
- magnesium.

Electrolyte products can be useful when:

- exercise is prolonged;
- sweat losses are high;
- training occurs in hot environments;
- substantial sodium and fluid replacement is needed.

Most short gym sessions do not automatically require an electrolyte drink.

Multivitamins

A multivitamin can help fill specific dietary gaps in some situations.

It does not replace:

- fruits;
- vegetables;
- protein-rich foods;
- whole grains;
- a balanced diet.

Very large doses of vitamins and minerals are not automatically beneficial and can be harmful.

Omega-3 Supplements

Omega-3 fatty acids are found naturally in fatty fish.

Plant foods such as:

- flaxseed;
- chia seeds;
- walnuts

provide alpha-linolenic acid.

Fish-oil and algae-derived supplements can provide long-chain omega-3 fatty acids.

Omega-3 supplements are not required for muscle gain or fat loss.

Pre-Workout Supplements

"Pre-workout" is a product category rather than a single nutrient.

Products can contain combinations of:

- caffeine;
- creatine;
- beta-alanine;
- citrulline;
- amino acids;
- vitamins;
- herbal compounds;
- sweeteners.

The effect depends on the actual ingredients and doses.

A product should not be assumed to work simply because it is labeled "pre-workout."

Mass Gainers

Mass gainers are high-calorie products usually containing:

- carbohydrate;
- protein;
- sometimes fat.

They can help people who struggle to consume enough calories through normal meals.

A homemade high-calorie shake can provide a similar function using foods such as:

- milk or soy milk;
- oats;

- banana;
- peanut butter;
- whey or plant protein.

Fat Burners

Most commercial fat-burner supplements do not produce substantial fat loss without a calorie deficit.

Many contain stimulants.

Possible problems include:

- increased heart rate;
- anxiety;
- disturbed sleep;
- gastrointestinal symptoms;
- uncertain ingredient combinations.

Fat burners are not necessary for fat loss.

Detox Products

Commercial detox teas, juice cleanses and similar products are not required to remove normal metabolic waste from the body.

Some products can cause:

- diarrhea;
- dehydration;
- electrolyte disturbance;
- excessive calorie restriction.

The liver and kidneys already perform major roles in processing and eliminating many waste products.